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Gamma Distribution Model Describes Maturational Curves for Delta Wave Amplitude, Cortical Metabolic Rate and Synaptic Density

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We analyzed the available ontogenetic data (birth to 30 years of age) for: amplitude of delta EEG (DA) waves during sleep; cortical metabolic rate (CMR) measured with positron emission tomography; and synaptic density (SD) in frontal cortex. Each is at the adult level at birth, increases to about twice this level by 3 years of age, and then gradually falls back to the adult level over the next two decades. Statistical analyses revealed that individual gamma distribution models fit each data set as well as did the best *ad hoc* polynomial. A test of whether a single gamma distribution model could describe all three data sets gave good results for DA and CMR but the fit was unsatisfactory for SD. However, because so few data were available for SD, this test was not conclusive.

We proposed the following model to account for these changes. First, cortical neurons are stimulted by birth to enter a proliferative state (PS) that creates many connections. Next, as a result of interactions in the PS, neurons are triggered into a transient organizational state (OS) in which they make enduring connections. The OS has a finite duration (minutes to years), and is characterized by high rates of information-processing and metabolism. Levels of CMR, SD and DA, therefore, are proportional to the number of neurons in the OS at any time. Thus, the cortex after birth duplicates, over a vastly greater time scale, the overproduction and regression of neural elements that occurs repeatedly in embryonic development.

Finally, we discussed the implications of post-natal brain changes for normal and abnormal brain function. Mental disorders that have their onset after puberty (notably schizophrenia and manic-depressive psychoses) might be caused by errors in these late maturational processes. In addition to age of onset, this neurodevelopmental hypothesis might explain several other puzzling features of these subtle disorders.

Introduction

One of us (Feinberg, 1982/83) has proposed that the human brain undergoes a profound maturational reorganization through late childhood and early adolescence, pointing to the following changes that take place over this period: (1) the decrease in brain metabolic rate; (2) the marked declines in the duration of deep (stage 4) sleep and in the amplitude of the delta (0-3 Hz) EEG that characterizes this stage; (3) the reduced brain plasticity, most clearly evident in the diminished ability to recover from brain lesions that cause aphasia, but also apparent in less drastic

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challenges to plasticity, such as the ability to learn a foreign language without an accent or to master complex psychomotor skills at high levels of skill (Feinberg & Carlson, 1968); (4) the reduction in latency of certain event-related potentials; and (5) the emergence of adult problem solving skills or cognitive "power".

In this paper, we argued that the reduction in cortical synaptic density that had recently been observed by Huttenlocher (1979) could account for each of these changes. This surprisingly late anatomical change is analogous to the "programmed cell death" that occurs in the fetus and may be the final manifestation of a mechanism used repeatedly in the early development of nervous systems: an initial overproduction of neural elements followed by a subsequent refinement or "pruning" [see Hamburger & Oppenheimer (1982); Purves (1988)]. This sequence of events could reduce by several orders of magnitude the genetic information required to construct the brain. Lastly, it was suggested that a fault in these late brain events might cause schizophrenia, whose onset in adolescence (especially in individuals who previously appeared entirely normal) has never been adequately explained.

At the time these hypotheses were advanced, evidence for a decline in brain metabolic rate over adolescence (Kennedy & Sokoloff, 1957), while statistically robust, was based on a small number of subjects in a limited age range. These data were obtained with the Kety-Schmidt method, which gives a weighted average for the rates of grey and white matter in the whole brain. Nevertheless, the decline in rate from children to young adults (of the order 25–30%) was not trivial, being equal to the average difference between normal and senile elderly. We noted in our review that metabolic studies with positron emission tomogrphy (PET) could substantially advance knowledge of this important physiological component of brain ontogeny: PET is less invasive than the nitrous oxide method and so can be applied to larger numbers of subjects. Moreover, PET can measure regional metabolism and thus distinguish ontogenetic changes in neocortex from those in other areas.

The results of such PET studies were recently reported by Chugani *et al.* (1987). The pattern they found for cortical metabolic rate (CMR)—a steep rise after birth to twice the adult levels, with a decline to these levels that begins in late childhood and is completed around the end of the second decade—was similar to that which had been desdribed for both delta wave amplitude (DA) (Feinberg *et al.*, 1977) and synaptic density (SD) (Huttenlocher, 1979). We used the new CMR data for frontal cortex (all neocortical areas were highly correlated with $r = 0{\cdot}98$ or greater) and the original DA and SD data to analyze statistically the curves of the three variables over the first three decades of life.

In examining plots of the raw data against age, we were struck by two features. First, the graphs for all three measures resembled the skewed outline of a gamma distribution. Secondly, it seemed that the curve for synaptic density would more closely resemble those for CMR and DA if a power of the SD data was used. We therefore examined the fit of each data set to a gamma distribution model, and used the square of synaptic density as the simple power function to compare with DA and CMR.

We first compared the fit of a gamma distribution model for each data set with that of the best-fitting empirical polynomial. Individual gamma models were fit to

DA, CMR, SD and SD squared using nonlinear least squares. The model is of the form:

$$y = d + a * e^{b*age} * age^{c}$$

where a, b, c and d are the model parameters.

Table 1 shows that the individual gamma distribution models for CMR, DA, SD, and SD squared fit the empirical data as well as the best *ad hoc* polynomial. Both accounted for remarkably high proportions of the variance. The individual gamma curves are shown in Fig. 1(a)-(d). Their similarity is striking, especially when one considers the widely disparate methods of measurement—synapse counts from EM micrographs, hand measurements of delta amplitude, and PET—used to obtain the data. This overriding similarity suggests the operation of a fundamental and pervasive process.

TABLE 1

Comparison of gamma distribution model and best polynomial

	Order of best polynomial	Fraction of explained Sum of squares	
		polynomial	gamma
Delta amplitude	3	0·676	0·670
Frontal cortex (CMR)	3	0·868	0·862
Synaptic density	4	0·841	0·925
Synaptic density^2	4	0·857	0·913

[The basis of the comparison is 1—(residual sum of squares)/(total sum of squares), which is equal to R^2 in a linear regression. None of the differences between gamma distribution and polynomial fits approached statistical significance.]

Figure 1 shows that the synaptic density curves peaked earlier than the curves for DA and CMR. The computed peak ages were 3·2, 6·1, and 7·4 years for SD, CMR and DA, respectively. The shapes of the three curves, while quite similar, were not identical. We reasoned that, if the age changes of DA and CMR are directly determined by synaptic density, the shape of the underlying gamma distribution model could be the same for all three variables, with the apparent differences due to measurement error. We therefore created a combined gamma distribution model, making parameters b and c (which determine shape) common to DA, CMR and SD squared, and allowing parameters d and a (which control position and scale) to vary for each data set. Since an earlier peak for SD was intuitively plausible (see below) DA and CMR were permitted to lag behind SD. (We recognize, however, that the location of the SD peak is uncertain because this curve is based on relatively few data points.)

Figure 2 shows the curves obtained. Table 2 shows that this combined model provided a good fit (compared to the individual gamma distribution models) for

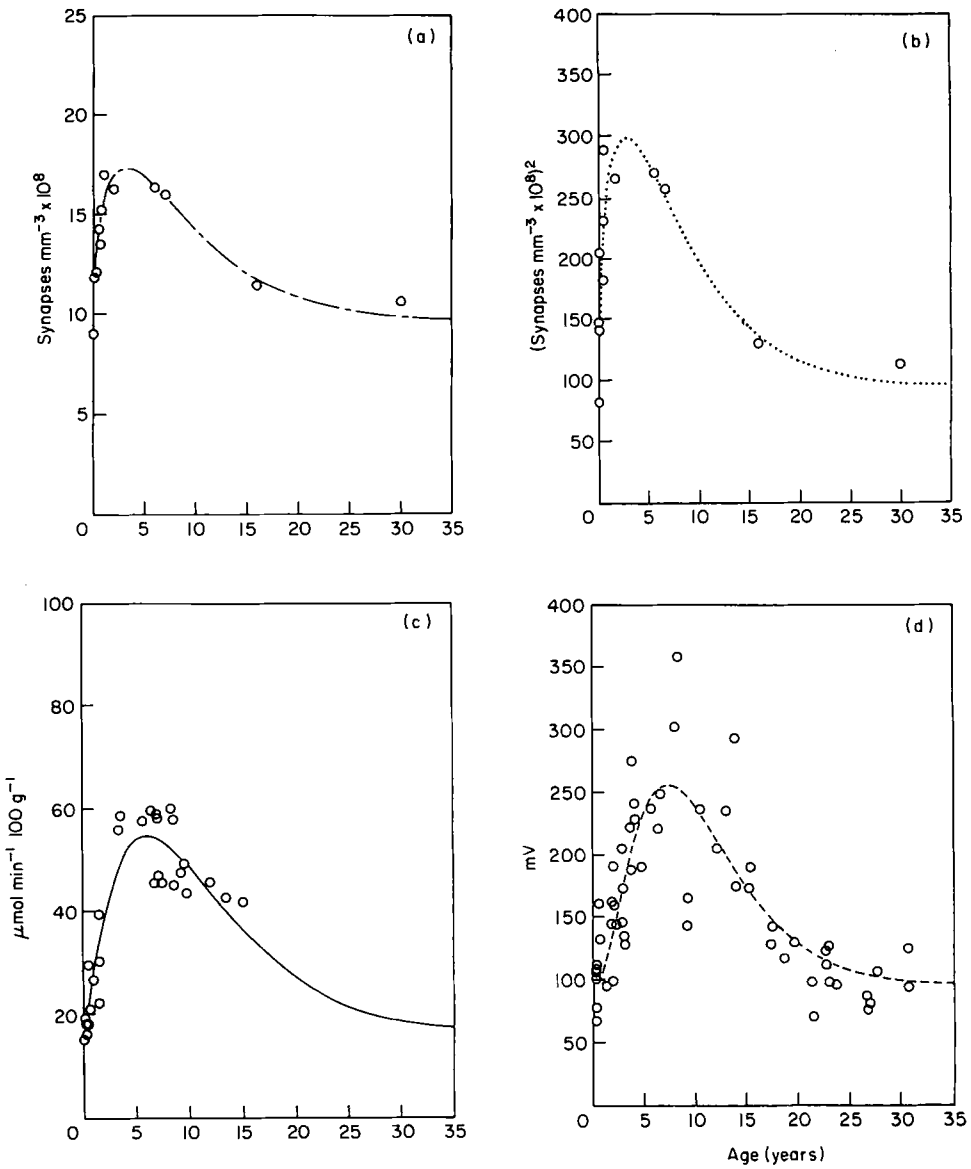


FIG. 1. (a)-(d) Individual gamma distribution models (curves) fitted to the raw data (scattergrams) for: synaptic density (a), synaptic density squared (b), cerebral metabolic rate (c), and delta amplitude (d). Only data from birth to 30 years of age were used. The curves for synaptic density peak earlier (3·2 years) than those for delta amplitude (7·4 years) and metabolic rate of frontal cortex (6·1 years).

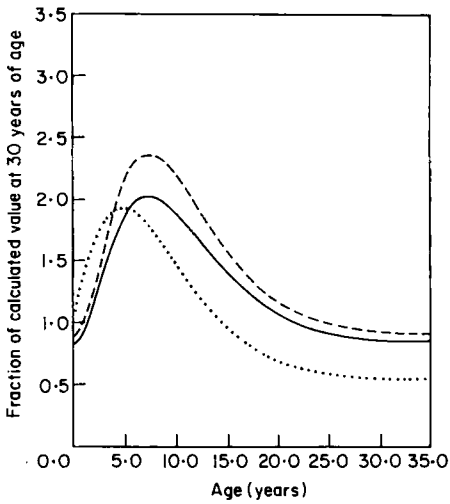


FIG. 2. Combined (common shape—see text) gamma distribution models for synaptic density squared (dots), delta wave amplitude (dashes) and cortical metabolic rate (solid line). Data were standardized to the value at 30 years of age for each measure. The curves for DA and CMR are virtually identical and account for nearly as much of the variance as the individual gamma models (Table 2). However, the curve for SD squared, which is based on the fewest subjects, accounts for substantially less variance than does its individual gamma distribution model.

TABLE 2

Comparison of individual gamma distribution models with combined model for each data set

	DA	CMR	Syn. den^2
N	57	29	12
R^2, individual model	0·67	0·86	0·91
R^2, combined model	0·67	0·79	0·67
Change in R^2	0·1%	8·2%	26·3%

DA and CMR but there was a substantial loss in R^2 for SD squared. The extent of the decrease in R^2 in the combined model for each variable was inversely correlated with the amount of data (i.e. the N) each contributed to the common shape parameters.

Obviously, more extensive data for synaptic density are needed for a definitive test of the hypothesis that a single gamma distribution model represents the ontogenetic curves for the three variables. The analyses here show only that the hypothesis is plausible. Nevertheless, our data indicate that individual gamma distribution models describe the maturational change for SD, DA and CMR at least as well as the best *ad hoc* polynomial and thus that the three sets of data exhibit a fundamentally similar age pattern.

In the remaining discussion, we first modify the overly simple description of the maturational reorganization we first proposed (Feinberg, 1982/83). Next, we consider the ways in which SD, CMR and DA might be interrelated. Lastly, we speculate further on the potential implication of late maturational changes for normal and abnormal brain function.

Initially, we emphasized the role of regressive changes in the hypothesized brain reorganization and we suggested that synaptic elimination could account for most of them. However, as Purves (1988) has cogently argued, regressive reorganizations generally include constructive components as well. To cite one of his examples: in peripheral systems (skeletal muscle, adendritic ganglion cells), when innervation by a single axon replaces the multiple axonal contacts present at birth, the synapses on the surviving axon become more dense and the axon's geometry more complex. Extrapolating Purves' analyses to the regressive central nervous system phenomena examined here, one would also expect them to have important constructive elements. These could include increases in synaptic strength and integration.

We now, therefore, conceptualize the post-natal maturation of the cerebral cortex as consisting of two stages, a proliferative state (PS) and an organizational state (OS). The PS lasts 1–3 years. It is triggered by birth, and includes the explosive growth of dendritic and axonal processes and the expansion of the neuropil illustrated in the Conel (1939–1963) atlases. The exuberant production of synapses suggested by Huttenlocher's data would be a logical component of the PS. We assume that the PS is genetically programmed, with connections determined by local chemoaffinities and relatively simple neural activity. The PS results in an extremely high level of neural connectivity that permits the competitive and other neuronal interations that will utilize information to construct complex, enduring circuits during the OS.

We assume that next, at some point after birth, a given neuron enters a transient "organization state" (OS); this state might be triggered by a particular temporal interaction with other neurons (which would be more likely to occur after it has established many synapses, even if relatively undeveloped). The neuron remains in this growth state for a finite period. After leaving this state, the neuron cannot return to it, perhaps because the state is produced by the one-time expression of a gene.† While in this state, the neuron is highly plastic and is highly active metabolically.

In its passage through the OS the neuron undergoes synaptic modifications that create long-lasting neural circuits. When the OS has been completed, most neurons have fewer synapses than when they entered. The levels of CMR and DA (over the levels at birth and maturity) are proportional to the number of neurons in the OS at any time. Thus, in this model, the number of neurons entering the OS increases during much of the first decade, partly as a consequence of an increasing number of active synapses.

† We recognize, of course, that considerable synaptic plasticity remains after virtually all neurons have left the OS. This plasticity is manifested by ongoing problem solving and the acquisition of new information and skills. There also exists sufficient plasticity to reorganize function after brain injury or altered input from the periphery (see e.g. Merzenich, 1985). But it also seems clear from several lines of evidence that there exists a unique form of plasticity in early life that is irretrievably lost around the onset of the third decade.

Considering the broad tendencies of the above process, we see that, as an increasing proportion of eligible neurons pass through this OS, the number of remaining, eligible neurons begins to decline. We then have fewer neurons actually in the OS, so that CMR and DA decline.

It is possible to model this hypothetical process quantitatively in several ways that are consistent with our observed data. For example, this gamma distribution could arise statistically as the proportion of neurons which are in the transient OS at each age, if we assume that a neuron enters the OS after several occurrences of some stimulus which is widely, but sparsely, distributed in the cortex both in location and in time (i.e. is distributed approximately randomly, from a macroscopic perspective.) Such a stimulus might be a particular input pattern from other neurons, so that the neuron would enter the OS only when that pattern was reinforced; or it might be the acquisition of a trophic factor.

We recognize that a quantitative model such as that above is overly simple. For example, it assumes a constant baseline level underlying the growth and decay of these curves. However, the fact that CMR, DA and SD levels at birth are roughly equal to those at maturity is almost certainly coincidental, given the vast differences between the neonatal and mature brain in structure and functional organization. As noted above, the brain, including the cortex, grows rapidly after birth, due to the extension and proliferation of neural processes and to myelination. Such growth must account for a substantial portion of the metabolic demand in the infant's brain. In contrast, CMR in the adult is largely concerned with sustaining membrane potential, and with the metabolic demands associated with action potentials and subthreshold post-synaptic changes. Similarly, while peak-to-peak delta wave amplitudes are roughly equal in the newborn and adult, the organization of the EEG is entirely different. With regard to SD, Huttenlocher points out that the morphology of synapses in the neonate differs from that at the end of the first year. Functionally, of course, the cortex has minimal capability of processing information at birth and full capability by the end of the second decade of life.

We turn now to the mechanisms by which the OS could produce high CMR and DA. A neuron would typically have entered the OS when its synapses are close to peak number; many which will later be lost are still present and active. Thus, the larger membrane surface area associated with higher synaptic density requires more energy for its maintenance (see Chugani *et al.*, 1987). Moreover, the higher level of interconnections will involve more neurons in any given activity, increasing overall CMR. Thirdly, we assume that both the regressive and constructive synaptic changes are metabolically demanding.

A relation of EEG amplitude to a high density of active synapses appears straightforward. EEG waves are believed to represent summed post-synaptic potentials in large assemblies of cortical neurons and dendrites being driven in synchrony by a subcortical "pacemaker" (Elul, 1972). Higher synaptic density could, therefore, produce larger EEG waves in two ways. First, the increased membrane surface (more dendrites/neuron) could produce a larger average voltage response/neuron to the synchronizing stimulus. Secondly, higher levels of interconnection would permit larger aggregations of neurons to change potential synchronously.

However, there is another maturational change in sleep, one closely correlated with that in DA (Feinberg *et al.*, 1989) that also must be explained. The child has a longer duration of deep sleep (more delta waves/min) as well as having delta waves of high amplitude. The increased delta wave duration might be understood as follows. We assume that one function of sleep is to recover from some of the metabolic consequences of information-processing during waking. If so, the intense information-processing of the child's brain, which guides the construction of enduring circuits, should require longer perods of deep sleep for recovery. [Following our original hypothesis (Feinberg, 1974), it is now generally accepted that it is primarily slow-wave or non-rapid eye movement sleep that is restorative for the brain and that the recovery process is most intense during high amplitude delta sleep.] A relation between the intensity of brain activity during waking and the duration of the "brain recovery" component of sleep would logically hold whether "brain recovery" is active (increased or qualitatively unique neuro-metabolic activity) or passive (a simple decrease in neuronal metabolism or "rest").

The model tentatively outlined above postulates that the OS is self-limited. We chose to make this assumption because it lent itself to the simple quantitative models we are exploring. However, there exist other plausible possibilities. The central physiological facts that require explanation are the late declines in CMR and DA.

One possible alternative to a self-limited OS is that this state is terminated by some external influence such as a non-central nervous system hormone. Initially, we thought that DA, CMR and SD started their declines around puberty and we suggested that this change might be triggered by sex hormones, perhaps acting to raise a threshold required for trophic maintenance of neural connections (Feinberg, 1982/83). This speculation no longer seems tenable because, as noted above, DA and CMR begin to decline around the middle of the first decade of life, and SD even earlier. [Extensive new computer measurements of DA recently carried out in our laboratory are consistent with a pre-pubertal onset of regressive changes: integrated amplitude of delta (0–3 Hz waves) fell by 19% from 5 to age 10 years of age (and by a further 30% between 10 and 20 years of age Feinberg *et al.*, 1989)].

A somewhat better endocrine candidate for influencing late regressive processes might be the adrenal androgens. A surge in adrenal androgen secretion begins in both sexes at about age 6 years of age (adrenarche). The levels of these hormones then increase steadily until a plateau is reached at about age 16 years of age (Reiter *et al.*, 1977). Thus, over the ages 6–16 years, the rise in adrenal androgens mirrors the fall in CMR and DA. However, it is clear that maturity in some complex neuronal systems (e.g. those serving ocular dominance) is achieved before adrenarche. Thus, if adrenal androgens stimulate neural regression, their effect would have to be on systems that mature later. Again, the speculative nature of this discussion is apparent from the fact that there is no evidence that these androgens affect brain regression. Perhaps some speculation in this area is permissible because no developmental role for these interesting hormones has yet been established.

One point we would make parenthetically here: neither the CMR nor the DA decline can be explained by a late decline in myelination. There is vastly more myelination underway at 1 year old than at 3 years of age (Yakovlev & Lecours,

1967), but both CMR and delta amplitude are higher at 3 years of age (Fig. 1). Moreover, Yakovlev & Lecours (1967), whose paper is usually cited as demonstrating significant myelination through the second decade, were quite cautious regarding the conclusions that could be supported by their data. They emphasized that, "Only a *few* cerebra suitable for normative study were obtained in the age-range of the first and second decades. This critical gap has limited our exploration of cycles of myelination particularly in the *telencephalic* and *cortical* regions" (Yakovlev & Lecours, 1967: 6, italics added).

Whatever the causal mechanism(s), it seems at least plausible that such extensive and protracted changes in synaptic density, cortical metabolic rate and delta wave amplitude would have correlates in altered brain function. In what follows, we extend our previous ideas regarding this possibility, especially with respect to the possibility that errors in late brain development might cause psychopathology. We emphasize the highly speculative nature of the following discussion, which we justify, in part, by the paucity of efforts to relate advances in basic neuroscience to the unsolved problems of mental illness.

In our previous proposal for a brain reorganization during adolescence, we hypothesized that elimination of redundant synapses may be required to achieve the sustained problem-solving capacity of the adult. Thus, late synaptic elimination may represent, in part, an exchange of relative equipotentiality for cognitive specialization and the maintenance of the extensive and complex neural circuitry required for sustained problem-solving. The possibility that maturation of cognitive function may require, as one component, elimination of "redundant" synapses, has recently been recognized by others (Goldman-Rakic, 1987).

Many children manifest transient aberrant behavior (obsessions, phobias, etc) that they "grow out of". The regressive/constructive fine-tuning during maturation may explain such improvement, a possibility implied by Hamburger & Oppenheim (1982). One fairly common syndrome with this pattern is childhood hyperactivity or attention-deficit syndrome (although not all children grow out of it). It seems intuitively plausible, as we noted previously, that "excessive" synapses could interfere with sustained attention.

Whether a fault in these late maturational brain changes plays a role in the development of psychiatric disorders is unknown. But a fundamental and unexplained fact is that *typical* mania, depression and schizophrenia are rare before puberty and common shortly thereafter (Feinberg, 1982/83; Feinberg, 1988). If one accepts the evidence that extensive brain changes persist through late childhood and adolescence, it seems possible that this temporal span includes a "critical period" when a fault in the process may cause or predispose to mental illness. Certainly, a precedent exists for this possibility within the central nervous system, where interference with normal developmental events in the first years of life (e.g. deprivation of binocular input) permanently impairs normal function and synaptic organization (e.g. stereoptic vision and the formation of ocular dominance columns in the striate cortex). The hypothesized developmental defects that causes mental illness might be stochastic, genetic, environmentally produced, or result from an interaction of these variables.

Purves (1988) has reviewed the role of trophic factors in maintaining connections between nerve and muscle and between neurons in peripheral ganglia. If such factors control synaptic organization within the central nervous system (an unproved but likely proposition, as Purves emphasizes) one would expect that controlling mechanisms might become more tenuous as more and more feedback and associative systems are interposed between primary sensory and motor centers.† In this way, more complex brains may become more susceptible to more subtle disorders. These subtle impairments may result in intermittent appearance of symptoms because of transient imbalances among large neuronal systems that are themselves intact.

The major functional psychoses (schizophrenia, manic-depressive disorders), are, in fact, extremely subtle illnesses although their behavioral expressions can be gross and extreme. Indeed, almost all brain functions (including most aspects of complex information-processing) can be normal in severely deranged individuals. The severity of the symptoms fluctuates over time. These psychoses have historically been called functional because it has not been possible to demonstrate any consistent abnormality—either physiological or morphological—in the brains of those afflicted, a fact that remains true today.

If the function of late refinements of brain connectivity is to ensure correct quantitative and qualitative synaptic organization of complex neural circuits, one could conceive (at least, metaphorically) of two different kinds of abnormality resulting from an error in these processes. In manic-depressive psychoses, the behavioral disorders often resembles extreme forms of normal emotional states. One is, therefore, tempted to think that the abnormality in this condition is a quantitative one.

The bizarre thinking of schizophrenia, on the other hand, might result from qualitatively incorrect neural circuitry. Elsewhere, one of us proposed that many of the abnormalities of thought processes in schizophrenia could be understood as a failure of feed-forward monitoring mechanisms (Feinberg, 1978). Such failure could impair the ability to distinguish between endogenous (internal images or thoughts) and externally induced (perceptions) brain activity. Difficulties in making this distinction are (at least, phenomenologically) a common problem for schizophrenic patients.

One of the vexing diagnostic problems in psychiatry is the existence of mixed syndromes, where both schizophrenic and affective symptoms are present. If both disorders stem from varying abnormalities in the same maturational processes, such mixtures would be less surprising. This interpretation could also explain the fact that schizophrenic patients can have a family history of major affective disorder and that the converse also occurs.

The possibility that late developmental brain changes play an etiologic role in schizophrenia has recently drawn the attention of other investigators. Weinberger

† In his book, Purves (1988) considers the importance of trophic mechanisms for the neuronal rearrangements that maintain appropriate innervation of a growing body. One wonders whether a converse dependence of body growth mechanisms on brain requirements might not also exist. Perhaps the prolonged juvenile state in man and other primates evolved to allow time for the brain to construct the neural circuits required in behavioral maturity. In this regard, it is of interest that monkeys also demonstrate post-natal synaptic proliferation and regression (O'Kusky & Colonnier, 1982; Rakic *et al.*, 1986).

(1987) and Murray & Lewis (1988) both proposed that schizophrenia might come about as the result of the interaction of late developmental changes with some pre-existing "lesion". We find this addition to the developmental hypothesis to be unparsimonious, because, as noted above, no histopathological lesion of the sort they envision has been demonstrated. Finch & Morgan (1987) speculated that an "asynchrony" in receptor "aging" might cause schizophrenia. Considering the age incidence of functional psychoses, one would think that "maturation" rather than "aging" is involved. This distinction is not entirely semantic since the two processes probably have quite different biological mechanisms.

Whether or not the late brain changes analyzed in this paper are significantly related to cognitive development or psychopathology remains, we reiterate, entirely conjectural. But these possibilities seem worth investigating in view of the magnitude of these changes and the dearth of clues to the etiology of the major psychoses.

Research into these questions could be facilitated by the fact that the ontogenetic curves for delta wave amplitude and cortical metabolic rate are virtually identical. This suggests that DA, which is easily measured by computer (Feinberg *et al.*, 1978), can provide a non-invasive and relatively inexpensive index of the underlying maturational process. The quantitative changes in delta EEG (especially of its duration which can be measured more precisely than its amplitude), might reflect the "magnitude" of the underlying brain events. Lastly, we note that Courchesne *et al.* (1987) recently found that the amplitude of the *Nc* event-related potential appears to parallel Huttenlocher's synaptic density curve as well. If their observations are confirmed and the variability of this measure is within acceptable limits, *Nc* amplitude might provide a still simpler and less expensive indicator for this ontogenetic pattern.

It also seems important in future research to confirm and expand the ontogenetic data for synaptic changes in human neocortex. Huttenlocher's pioneering study was understandably limited in the number of suitable brain specimens that could be acquired and studied by a single investigator. Although his findings were published a decade ago, a recent search failed to reveal any new data on synaptic density in human frontal or association cortex.

However, Huttenlocher & de Courten (1987) recently published further data for synaptic density in human visual cortex. This structure, being delimited by the line of Gennari, has the advantage that one can be certain that the same cortical tissue is being sampled at different ages. Huttenlocher & de Courten found that, synapses in visual cortex show a pattern of post-natal growth and regression similar to that of frontal cortex but that the time course is different: both the peak density and the adult level are reached earlier (8 months and about 11 years, respectively), as might be expected for a structure that matures earlier and, presumably, has a higher proportion of "hard wiring" than frontal or association cortex.

Magnetic resonance imaging (MRI) has recently provided further, independent evidence that morphologic changes in cortex persist into the second decade of life. Jernigan & Tallal (1990), using a method of automated analysis, found substantial reductions in cortical volume between children and young adults. In a more recent study (T. L. Jernigan *et al.*, personal communication), they show that the decline

is linear between 8-30 years of age ($r = -0{\cdot}78$, $p < 0{\cdot}001$) and that it is maximal in the frontal and parietal convexities.

Whether or not Huttenlocher's data for synaptic density in frontal cortex are confirmed in detail, the ontogenetic patterns for delta wave amplitude and cortical metabolic rate, and Jernigan and coworkers' new MRI data for cortical changes in cortical volume, (each of which is based on relatively large numbers of subjects), provide strong and independent evidence that profound maturational changes persist in the human brain through the second decade of life. Exploration of the functional significance of these changes could provide a challenging arena for the study of brain-behavior relations.

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